

WILDEYE: REAL-TIME WILDLIFE INTRUSION DETECTION AND DETERRENT SYSTEM USING AI AND IOT

PROJECT REPORT

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To

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Bachelor of Technology in Computer Science and Engineering



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SCMS SCHOOL OF ENGINEERING AND TECHNOLOGY
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DECLARATION

We undersigned hereby declare that the project report **WILDEYE: REAL-TIME WILDLIFE INTRUSION DETECTION AND DETERRENT SYSTEM USING AI AND IOT**, submitted for partial fulfilment of the requirements for the award of the degree of Bachelor of Technology (CSE) of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of Ms. Doney Daniel. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the sources. We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma or similar title of any other University.

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled **WILDEYE: REAL-TIME WILDLIFE INTRUSION DETECTION AND DETERRENT SYSTEM USING AI AND IOT** submitted by N M SREEHARI [SCM22CS162], PRANAV VARMA [SCM22CS166], PRINCE JACOB BOBBY [SCM22CS168], RAHUL K R [SCM22CS170] to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology is a bonafide record of work carried out under our guidance and supervision during the academic year 2025-26.

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ABSTRACT

Human-wildlife conflict remains a significant challenge in agricultural regions worldwide, leading to substantial crop damage, economic losses for farmers and threats to both human and animal safety. Traditional deterrent methods, such as physical fencing and manual patrols, are often labor-intensive, costly and ineffective over large areas. Recent advancements in AI and IoT offer a transformative approach for automated, real-time monitoring and response systems. This project presents an intelligent system for the real-time detection and deterrence of wildlife intrusions. The system leverages a Raspberry Pi 4 integrated with a camera module and motion sensors to monitor farmland. A custom-trained YOLOv8 deep learning model is employed for accurate animal detection and species classification. Upon identification, the system activates a non-invasive, species-specific deterrent mechanism that utilizes multi-sensory repellents, including specific light patterns, acoustic signals and odor emitters, informed by indigenous knowledge. An IoT framework facilitates real-time alerts and remote control via a mobile application and Telegram bot, enabling user feedback and system interaction. Experimental results validate the system's efficacy in reliable intrusion detection and classification. Overall, this project demonstrates the potential of integrated AI-IoT solutions in promoting sustainable agriculture, enhancing farmer security and fostering human-wildlife coexistence.

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ABBREVIATIONS

AI	Artificial Intelligence
DL	Deep Learning
ML	Machine Learning
CNN	Convolutional Neural Network
IoT	Internet Of Things
YOLO	You Only Look Once
DCNN	Deep Convolutional Neural Network
mAP	Mean Average Precision
IoU	Intersection Over Union
ROI	Regions Of Interest
NCNN	Neural Convolutional Network Framework

CHAPTER 1

INTRODUCTION

1.1 Overview

Human-wildlife conflict represents a significant and growing challenge in agricultural regions globally, causing substantial crop damage, economic losses for farming communities and posing risks to both human safety and wildlife conservation. In many vulnerable areas, wildlife intrusions can lead to crop losses exceeding 50%, severely threatening food security and rural livelihoods. Conventional protection methods, including physical barriers, manual patrols and auditory deterrents, typically prove inadequate due to their high labor requirements, limited scalability and decreasing effectiveness as animals become habituated. These shortcomings underscore the critical need for an intelligent, automated solution capable of providing timely and targeted intervention[1].

The convergence of Artificial Intelligence and Internet of Things technologies has created new opportunities for developing sophisticated monitoring and response systems. Deep learning architectures, particularly convolutional neural networks and specialized object detection models like YOLO, have demonstrated remarkable performance in real-time visual recognition tasks. When deployed on edge computing platforms such as single-board computers, these models enable local processing capabilities without dependence on continuous network connectivity. Concurrently, IoT frameworks facilitate seamless integration between sensing devices, processing units and actuator systems, enabling comprehensive ecosystem management[5].

This project develops an integrated system that harnesses these technological advancements to address wildlife intrusion challenges. The implemented solution combines motion sensors and visual monitoring components with an optimized deep learning model for accurate animal detection and classification. Upon identifying specific species, the system activates tailored, non-invasive deterrent mechanisms that leverage multi-sensory approaches—including visual, auditory and olfactory stimuli—informed by ecological behavior patterns. The architecture incorporates remote monitoring and control capabilities through mobile interfaces, enabling real-time alerts and facilitating continuous system improvement. By providing an automated, adaptive and humane protection system, this work aims to enhance agricultural security while promoting sustainable coexistence between human activities and wildlife populations[1][2].

1.2 Problem Statement

Human-wildlife conflict in agricultural regions primarily relies on traditional mitigation methods such as physical fencing and manual patrols, which are often labor-intensive, costly and ineffective over large areas. These conventional approaches lack real-time responsiveness and fail to provide species-specific deterrence, leading to continued crop damage and economic losses for farmers. Furthermore, the absence of automated monitoring and intelligent response mechanisms results in delayed interventions and potential harm to both wildlife and human populations. While recent advancements in deep learning and IoT offer potential solutions, individual models often struggle with challenges such as varying environmental conditions, animal camouflage and computational constraints on edge devices. Therefore, there is a critical need for an integrated, intelligent system that can automatically detect wildlife intrusions in real-time, accurately classify animal species and activate appropriate, non-invasive deterrents. This project addresses these challenges by developing a robust system that combines optimized deep learning models with IoT-based sensing and actuation to provide a reliable and scalable solution for preventing crop damage while promoting wildlife conservation.

While many modern AI solutions exist, they often rely on high-bandwidth cloud processing for image recognition. In the remote agricultural belts where human-wildlife conflict is most severe, internet connectivity is often unstable or non-existent. This creates a dangerous "latency gap," where an intrusion is detected but the alert or response is delayed due to network failure. A viable solution must prioritize Edge Computing, performing all heavy deep-learning inference locally on the hardware to ensure an immediate, split-second response that does not depend on an external server.

Traditional motion-based security systems are frequently triggered by "false positives," such as swaying vegetation, falling branches, or domestic livestock. In a farm environment, wild animals are experts at utilizing natural cover, making them difficult to distinguish from the background using standard sensors. Relying on simple PIR or motion detection alone is insufficient. There is a requirement for Advanced Computer Vision that can extract deep features—such as specific body textures and gait patterns—to accurately classify wildlife even when they are partially obscured or camouflaged against dense foliage.

1.3 Objective

The primary objective of this project is to design and implement an automated real-time wildlife intrusion detection and deterrent system using deep learning and IoT technologies. The project aims to develop an optimized YOLOv8 model capable of accurately detecting and classifying common animal species in agricultural settings. It further seeks to estab-

lish a responsive IoT framework that integrates motion sensors, camera modules and multi-sensory deterrent mechanisms to enable immediate, species-specific intervention. Additionally, the project aims to create a user-friendly mobile and web interface that allows farmers to monitor alerts, provide feedback and manually control deterrents remotely. By combining efficient edge computing, reliable animal classification and non-invasive repellent techniques, the system strives to reduce crop damage, minimize human-wildlife conflict and support sustainable agricultural practices[5].

The project aims to engineer a lightweight yet robust deep learning architecture based on the YOLOv8n variant. The primary goal is to achieve high Mean Average Precision while maintaining a low computational footprint, enabling real-time inference directly on the Raspberry Pi 4. This includes optimizing the model to handle diverse environmental variables such as varying lux levels, motion blur, and partial animal occlusion, ensuring that detection remains reliable under "wild" field conditions.

The overarching mission of this work is to provide a Humane Alternative to traditional, often lethal, wildlife control methods. By focusing on non-invasive repulsion rather than physical exclusion or harm, the project aims to minimize crop losses and economic distress for farmers while simultaneously supporting local biodiversity and reducing the risk of fatal human-animal encounters.

CHAPTER 2

LITERATURE SURVEY

Several research studies have focused on applying AI and IoT technologies to address the challenges of human-wildlife conflict and agricultural monitoring. These works explore various approaches, including the use of deep learning for animal detection, the deployment of sensor networks for intrusion alerts and the design of automated deterrent systems to prevent crop damage. While some studies emphasize the development of accurate, real-time object detection models suitable for edge devices, others investigate the integration of multi-sensory repellent mechanisms and IoT frameworks for remote system management and data analytics. A brief summary of the most relevant works is presented below:

1. IoT and AI-driven solutions for human-wildlife conflict: Advancing sustainable agriculture and biodiversity conservation [1]

Niloofer Abed et al. developed an integrated system leveraging YOLOv8 object detection alongside IoT sensor networks for sustainable agriculture. Their approach focused on real-time identification and immediate deterrence of wildlife intrusions, helping prevent crop damage and facilitate coexistence between farmers and local fauna. Field deployment demonstrated the solution's ability to rapidly classify animal species and activate tailored repellent mechanisms—such as lights or sounds—based on behavioral profiles, minimizing harm and habituation. By automating detection and response, the authors offered a framework that not only benefits food security and rural livelihoods but also advances ecological stewardship in agricultural communities.

The technical implementation utilizes a custom-trained YOLOv8 model designed to perform high-speed inference on resource-constrained hardware. By curating a specialized image dataset of five prominent wildlife species, the researchers achieved superior detection rates even in complex rural environments. The system integrates an adaptive feedback loop via a mobile interface, allowing farmers to monitor real-time analytics, provide manual corrections to the AI's classifications, and remotely override automated deterrents. This human-in-the-loop design ensures that the system evolves over time, improving its accuracy through continuous user interaction.

A significant contribution of this work is the optimization of the AI-IoT stack for device-level inference on Raspberry Pi hardware. This architectural choice ensures that the system remains cost-effective and functional in remote areas where high-bandwidth cloud connectivity is often unavailable. The hardware configuration was tested under varied lighting

and environmental conditions, proving that edge computing can handle the computational load of modern object detection without significant latency. This robustness is critical for "always-on" monitoring systems that must respond to animal intrusions within seconds to be effective.

Extensive field experiments validated that the system leads to a significant reduction in both crop loss and manual labor. Beyond immediate protection, the IoT backbone enables continuous data logging of wildlife movement patterns, providing a longitudinal dataset that supports evidence-based policymaking for regional wildlife conflict mitigation. In conclusion, the study demonstrates a scalable, animal-friendly approach to intrusion prevention, highlighting how localized AI can bridge the gap between agricultural productivity and biodiversity conservation.

2. Edge AI in Sustainable Farming: Deep Learning-Driven IoT Framework to Safeguard Crops from Wildlife Threats [2]

Konkala Venkateswarlu Reddy et al. presented a cutting-edge edge computing architecture employing EvoNet and a network of IoT sensors to secure farmlands from wildlife threats. Their research targeted low-connectivity environments, prioritizing decentralized on-device inference for faster response times and lower operating costs. The EvoNet detector achieved a 91.7% accuracy rate and immediately alerts farmers upon wildlife detection. These alerts enable real-time interventions—such as activating alarms or deploying sensory-specific deterrents—to prevent significant crop loss and maintain farm security.

A primary technical innovation of this framework is the use of Separable Conv2D layers within the EvoNet architecture. By decomposing standard convolutions into depthwise and pointwise layers, the model drastically reduces the number of mathematical multiplications required without sacrificing feature representation. This computational efficiency is critical for edge deployment on low-power IoT devices, allowing the system to maintain high frame rates for video processing while minimizing thermal throttling and energy consumption in the field.

The model further leverages transfer learning by incorporating pre-trained weights from EfficientNet-B0. This allows the system to utilize generalized pattern recognition capabilities learned from massive datasets, which are then fine-tuned for the specific visual characteristics of local wildlife. Additionally, the researchers integrated a spatial attention mechanism that forces the network to focus on relevant pixels containing the target animal while suppressing irrelevant background noise, such as moving vegetation or shadows. This attention-driven approach significantly improves classification performance in varied and visually cluttered agricultural environments.

The authors demonstrated that their decentralized AI framework lowers the dependence

on costly manual patrolling and traditional physical fencing, making it highly practical for large-scale and remote farming operations. The architecture supports modular scalability, allowing new detection models and specialized sensors to be integrated into the existing network as farm requirements evolve. Overall, this work illustrates how combining lightweight deep learning models with granular IoT deployment transforms modern farming efficiency, making precision wildlife management both accessible and sustainable for rural communities.

3. Novel Animal Detection System: Cascaded YOLOv8 with Adaptive Preprocessing and Feature Extraction [3]

Johnwesily Chappidi et al. introduced a multi-stage, cascaded model where YOLOv8 is enhanced through adaptive preprocessing techniques and refined feature extraction, significantly improving wildlife detection under challenging field conditions. The system's unique cascading architecture allows initial model layers to quickly filter and propose candidate objects, with subsequent layers extracting and refining features for robust classification. This approach yielded a 97% detection accuracy, even in visually complex scenes characterized by dense background vegetation or varied lighting conditions.

To address the common issue of image degradation in outdoor environments, the system employs Adaptive Histogram Equalization (AHE). This preprocessing technique enhances image contrast locally rather than globally, which is essential for identifying camouflaged animals or subjects in shadow. By stabilizing the input data before it reaches the neural network, the researchers ensured that the feature extraction layers could consistently identify edge gradients and textures that define specific animal species, regardless of the time of day or atmospheric interference.

The core innovation lies in the transition from a standard single-stage detector to an optimal Cascaded YOLOv8 model. Unlike traditional architectures that attempt to perform localization and classification in one pass, this cascaded approach decomposes the process into iterative stages. The initial stage identifies broad Regions of Interest (ROIs), which are then passed to specialized secondary layers for finer localization and high-resolution classification. This hierarchical filtering reduces false positives by allowing the model to "double-check" ambiguous detections before triggering a farm alert.

The research further demonstrated substantial improvements through anchor box optimization and dynamic sensor fusion strategies, ensuring reliable real-time monitoring across multiple species. By fine-tuning the anchor boxes to match the specific aspect ratios of local wildlife, the system achieves superior bounding box regression compared to generic models.

4. The IoT-based real-time image processing for animal recognition and classification using deep convolutional neural network (DCNN) [4]

Surya T et al. engineered a real-time animal classification system powered by the AlexNet DCNN architecture, embedded in an IoT network for rapid response. Their pipeline enables live streaming from camera modules deployed on farms, where images are instantly evaluated by the DCNN to identify intruding species with a reported accuracy of 95%. The system triggers automated alerts and response mechanisms—such as early warning systems—to mitigate human-wildlife conflict and prevent the disruption of local ecosystems.

The technical core of the framework utilizes the AlexNet architecture, which consists of multiple convolutional layers designed to extract intricate spatial features from raw pixel data. This deep learning approach allows the model to distinguish between various animal species even under challenging environmental conditions, such as low light or partial occlusion. By offloading heavy computational tasks to a centralized cloud or edge server within the IoT framework, the researchers ensured that classification occurs in near real-time, effectively eliminating the latency typical of manual monitoring.

Beyond software, the integration of specialized IoT hardware plays a critical role in the system's field-readiness. The study details a deployment strategy involving low-power camera nodes that remain in a standby state until motion is detected, thereby conserving energy—a vital requirement for remote agricultural sites lacking consistent power grids. Once triggered, high-speed data transmission protocols ensure that the captured imagery reaches the DCNN classifier and returns a command to local actuators, such as sirens or light deterrents, within seconds.

Experiments confirmed that this IoT-based automation significantly reduces farmer workload, lowers response time, and enables finer monitoring of field conditions across large-scale operations. The modular architecture supports scaling across diverse terrains and farm sizes, ensuring the framework's applicability beyond small-scale pilot deployments. Ultimately, the study's findings underline how the synergy between deep learning and IoT drives sustainable agriculture management by delivering actionable intelligence and swift interventions against wildlife-induced crop damage.

5. AI-enabled smart agriculture using IoT for sustainable crop production and wildlife monitoring [5]

S. K. Sahu, P. R. Singh, and M. P. Rout advanced a smart farming ecosystem driven by deep learning models and IoT infrastructure for crop monitoring and wildlife management. Their work integrated distributed sensor networks that collected data on plant health, soil metrics, and animal presence, which was processed using sophisticated AI algorithms for

real-time analysis and decision support. The system delivered precise recommendations for resource allocation and automated responses to detected threats, promoting both yield optimization and ecological balance.

The architecture emphasizes a multimodal data fusion approach, where diverse inputs such as soil moisture, ambient temperature, and visual data are synthesized to provide a holistic view of the farm environment. By utilizing deep learning architectures specifically optimized for heterogeneous data, the system can predict crop stress patterns and differentiate between benign local fauna and potential predatory threats. This predictive capability allows for pre-emptive resource management, such as adjusting irrigation schedules or activating non-invasive deterrents before significant damage occurs.

Furthermore, the research highlights the integration of an intelligent decision-support layer that operates between the IoT perception layer and the physical actuators. This middle-ware utilizes edge computing to process time-sensitive data locally, ensuring that critical alerts regarding wildlife intrusion are broadcasted instantly without relying on constant cloud connectivity. Such a decentralized processing model is particularly effective in rural agricultural zones where bandwidth may be limited, ensuring that the smart agriculture stack remains robust and responsive under varying network conditions.

The authors showcased significant improvements in crop productivity and resource efficiency while simultaneously ensuring wildlife monitoring for habitat preservation. The combined AI-IoT stack automated much of the farm management, reducing human error and facilitating adaptive decision-making for long-term environmental sustainability. Ultimately, the research underscores the vital role of AI-driven IoT platforms in transitioning traditional agriculture toward smarter, scalable, and greener practices that harmonize food production with wildlife conservation.

CHAPTER 3

SYSTEM ARCHITECTURE & METHODOLOGY

3.1 Dataset

3.1.1 Dataset Source

The dataset used for model development in this project was compiled from multiple publicly available sources to ensure diversity and robustness. The primary sources include Kaggle and Imagescv, which host extensive collections of wildlife images. The dataset is specifically curated to represent common animal species responsible for crop intrusions in Indian agricultural contexts, such as wild boars, elephants, deer and monkeys.

Regarding data quality, the dataset presented significant challenges common to real-world wildlife imagery. The original images varied greatly in terms of resolution, lighting conditions and the presence of complex, cluttered backgrounds.

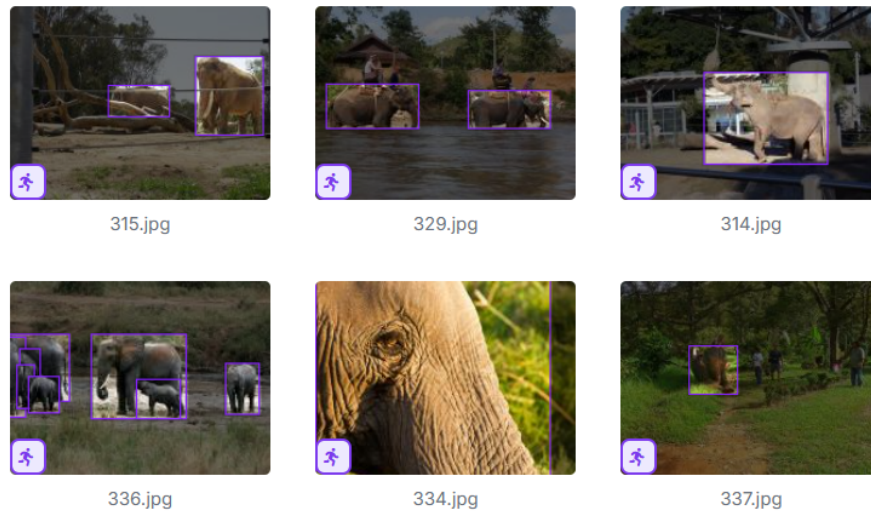


Figure 3.1: Dataset of Sri Lankan Elephants[12]

The final dataset was meticulously labeled using the YOLO PyTorch format, where each animal is enclosed in a normalized bounding box defined by class ID and coordinates (x, y, w, h) . Special attention was paid to "occlusion" cases—where animals are partially hidden by trees

or fences—ensuring the model learns to identify species based on partial visual cues. This level of granular labeling is critical for reducing false negatives in the dense vegetation common to Indian farmlands.

A critical aspect of wildlife monitoring is the transition between day and night. Since many crop-raiding animals, such as wild boars, are primarily nocturnal, the dataset includes a balanced mix of Daylight, Twilight, and Low-Light IR (Infrared) imagery. This ensures that the YOLOv8 model recognizes the specific silhouettes and heat signatures of animals even when traditional color information is limited, which is vital for a 24/7 security solution.

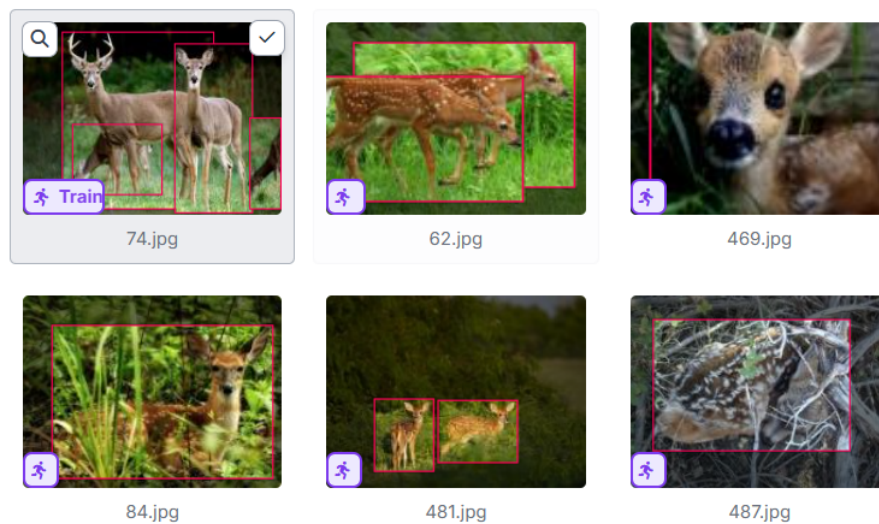


Figure 3.2: Dataset of Deer[14]

3.1.2 Data Preprocessing

Before training the YOLOv8 deep learning model for wildlife detection, a comprehensive preprocessing and augmentation pipeline was implemented using the Roboflow platform. This process was essential to standardize the dataset, enhance model performance and improve its ability to generalize to real-world conditions in agricultural settings.

All wildlife images were first auto-oriented to correct for incorrect rotation data, ensuring a consistent base orientation. The images were then resized using a stretch method to a uniform dimension of 640x640 pixels, which is the standard input size for the YOLOv8 architecture. This resizing ensures computational efficiency during training and inference while preserving the requirements of the model. To further standardize the input and potentially reduce model complexity, all images were converted to a single-channel grayscale format. This step helps the model focus on structural and textural features rather than color variations, which can be highly inconsistent in outdoor environments[3].

To significantly increase the diversity of the training data and prevent overfitting, a set of data augmentation techniques was applied, generating two new variants for each original training example. A key augmentation was the random adjustment of Exposure, varying between -50% and +50%. This technique simulates a wide range of lighting conditions—from bright daylight to deep twilight—thereby making the model robust to the challenging and variable illumination encountered in farmland throughout the day and night.

Overall, this preprocessing and augmentation phase transformed the raw collection of images into a standardized, balanced and varied dataset. This formed a solid and robust foundation for training a reliable deep learning-based wildlife detection system capable of performing under diverse and unpredictable environmental conditions.

3.2 System Diagrams

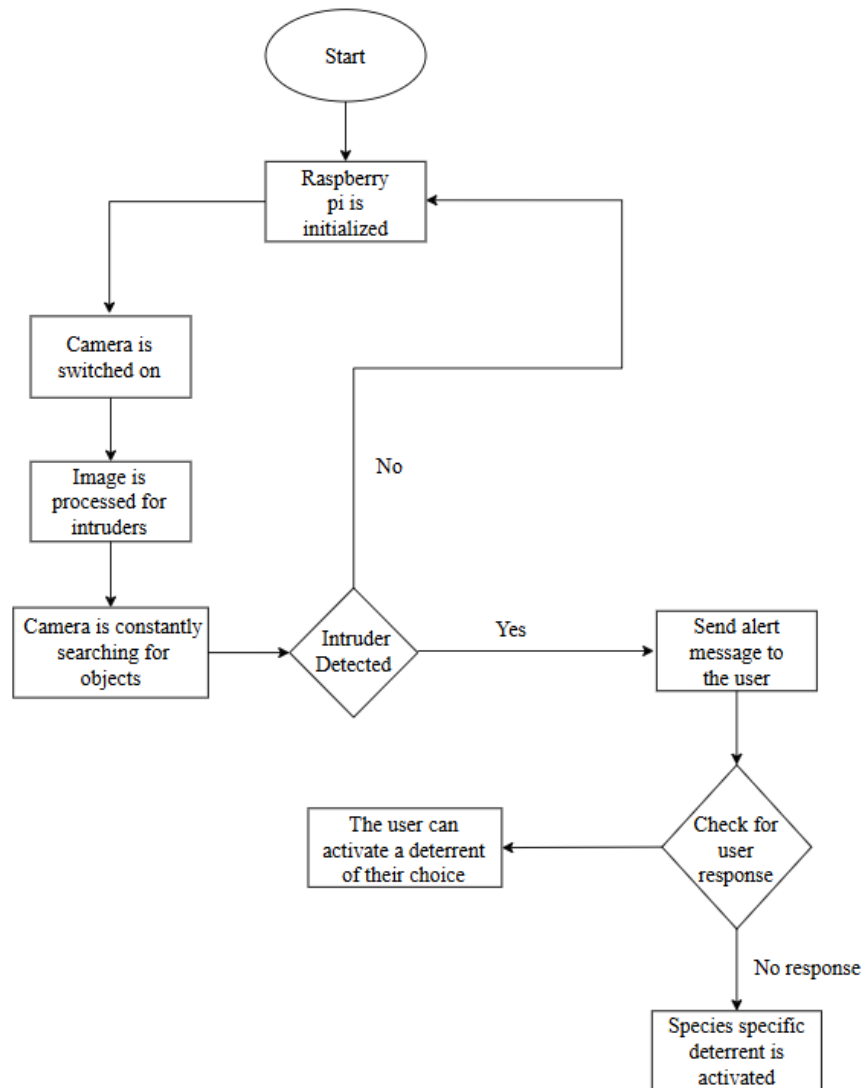


Figure 3.3: System Flow Diagram

3.3 System Architecture

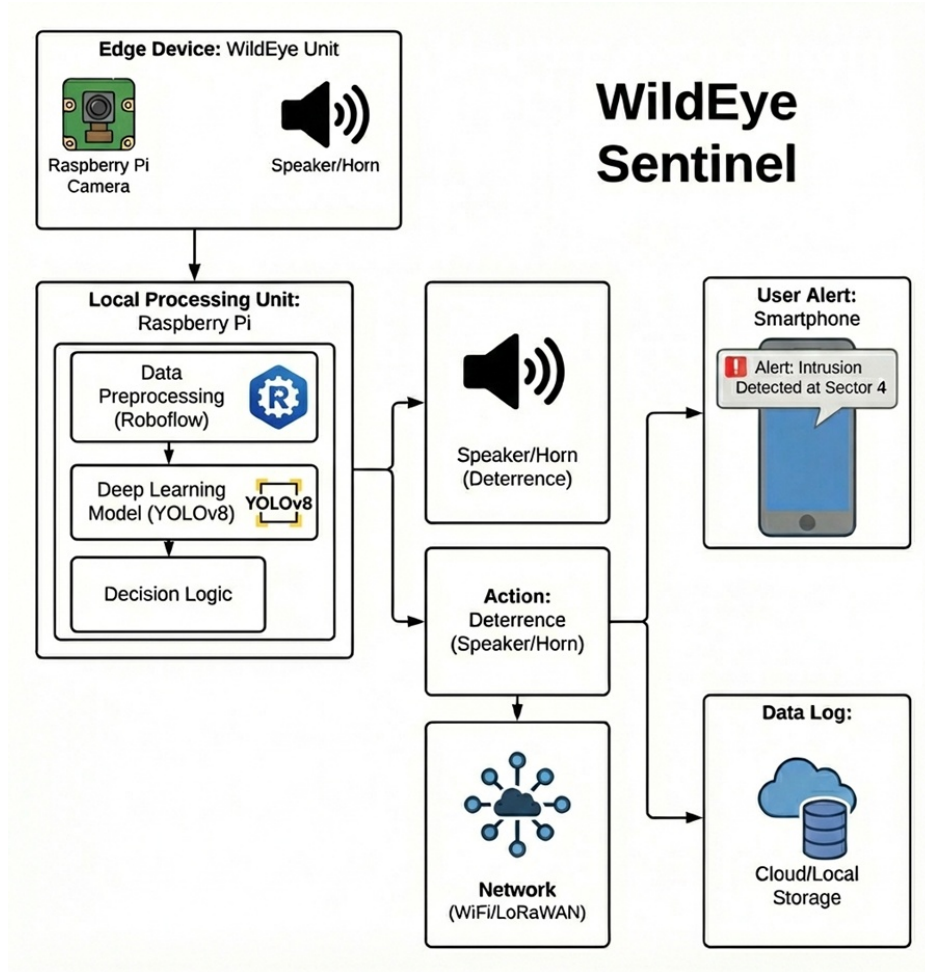


Figure 3.4: System Architecture

3.4 Methodology

3.4.1 System Overview

The proposed system integrates Artificial Intelligence and Internet of Things technologies to provide a real-time, automated solution for wildlife intrusion detection and deterrence. The entire system is controlled by a Raspberry Pi 4, which serves as both the central processing unit and control hub. It connects to sensors, a camera module and deterrent devices. The Raspberry Pi continuously monitors the field for motion, classifies detected animals and triggers appropriate deterrents based on the animal type[2][4].

3.4.2 Animal Detection and Classification

A custom-trained YOLOv8 model is deployed on the Raspberry Pi to detect and classify wildlife species. The model is trained using a curated dataset of native animals such as elephants, boars and deer[4]. Preprocessing steps such as data augmentation, resizing and normalization are applied to improve performance and robustness. During real-time operation, when the camera captures an image after motion detection, the model identifies the animal species. Performance metrics such as mAP, IoU and F1-score are used to evaluate detection accuracy[3].

3.4.3 Sensor Integration and Control System

The Raspberry Pi interfaces with an PIR sensor to detect motion within a defined range. This sensor acts as the trigger for the system. When movement is detected, the Raspberry Pi activates the camera module to capture an image. This tiered activation approach helps conserve power by ensuring the image processing module is only engaged when necessary. The embedded control module on the Raspberry Pi then manages the subsequent image classification workflow. The system can be used in two ways, allowing it to function either as a power-efficient PIR-triggered device for targeted alerts or as a comprehensive continuous surveillance station for real-time monitoring[6].

3.4.4 IoT and Feedback Mechanism

The system includes a Telegram IoT bot for remote user interaction and real-time alerts. Following the classification of a detected animal, the Raspberry Pi automatically sends a notification to a predefined Telegram user or channel. This alert includes the captured image annotated with the bounding box and the classification label, providing immediate visual confirmation of the intrusion. This mechanism also allows users to provide feedback on the classification, creating a valuable data loop for future model refinement[6].

3.5 Module Description

3.5.1 Hardware

Raspberry Pi 4

The Raspberry Pi 4 Model B (4GB RAM) serves as the high-performance central processing unit for this project, offering a significant leap in computational capability over its predecessors. Powered by a Broadcom BCM2711 quad-core Cortex-A72 64-bit processor running at 1.5GHz, it provides the necessary speed to handle complex tasks like real-time image

processing and neural network inference. The 4GB of LPDDR4 RAM ensures smooth multitasking and provides a sufficient memory buffer for loading the YOLOv8n model weights and processing high-resolution video frames without bottlenecking the system.

In this deployment, the Pi 4 acts as an Edge Computing gateway, managing the hardware-software interface between the ultrasonic sensors, the camera module, and the alerting system. Its support for dual-band 802.11ac wireless networking and Bluetooth 5.0 allows for seamless integration with the Telegram Bot API for remote notifications.

IR-Cut Camera

The OV5647 5MP Camera Module serves as the visual sensor for the wildlife detection system. It is equipped with a 1/4-inch CMOS sensor capable of capturing high-definition 1080p video at 30fps and 5-megapixel static images. A defining feature of this specific module is the integrated IR-Cut filter, which addresses the common "color distortion" issue found in standard night-vision cameras. During daylight, the filter is automatically engaged to block infrared light, ensuring natural color reproduction and sharp contrast—essential for the YOLOv8n model to accurately distinguish between animals and their backgrounds.



Figure 3.5: Raspberry Pi with Camera Module

3.5.2 YOLOv8

YOLOv8n (YOLOv8 Nano) represents the most compact and efficient iteration of the YOLOv8 family developed by Ultralytics, specifically engineered for real-time inference on edge devices with limited computational resources. As a one-stage object detector, it achieves high throughput by utilizing an anchor-free architecture that predicts object centers directly, which reduces complexity compared to previous anchor-based generations. The model incorporates a C2f (Cross-Stage Partial Bottleneck with two convolutions) module to enhance feature extraction and gradient flow, alongside a decoupled head that separates the classification and regression tasks for improved accuracy. With approximately 3.2 million parameters, YOLOv8n balances a remarkably small memory footprint with high-speed performance, making it the industry standard for applications requiring rapid detection on mobile processors, drones, and embedded systems without sacrificing the sophisticated data augmentation and architectural improvements of the full YOLOv8 suite[11].

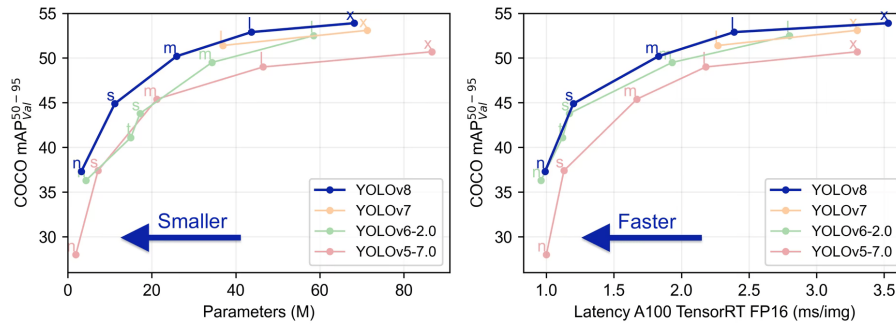


Figure 3.6: Comparison of YOLOv models[11]

Key Features of YOLOv8

- **Advanced –and Neck Architectures:** YOLOv8 employs state-of-the-art backbone and neck architectures, resulting in improved feature extraction and object detection performance[11].
- **Anchor-free Split Ultralytics Head:** YOLOv8 adopts an anchor-free split Ultralytics head, which contributes to better accuracy and a more efficient detection process compared to anchor-based approaches[11].
- **Optimized Accuracy-Speed Tradeoff:** With a focus on maintaining an optimal balance between accuracy and speed, YOLOv8 is suitable for real-time object detection tasks in diverse application areas[11].
- **Variety of Pretrained Models:** YOLOv8 offers a range of pretrained models to cater to various tasks and performance requirements, making it easier to find the right model for your specific use case[11].

Comparative Performance Analysis

1. **Accuracy and Convergence:** The training curve demonstrates a clear hierarchy in learning efficiency. The Tuned YOLOv8n exhibits a “steep learning curve,” achieving over 90% accuracy within the first 20 epochs. This suggests that the optimized learning rate and augmentation policies allowed the model to extract features more rapidly. In contrast, AlexNet shows significant volatility and a much lower “ceiling,” plateauing near 65%, highlighting the architectural limitations of older CNNs.
2. **Class-Specific Precision:** Precision measures the “trustworthiness” of a positive prediction.
 - **The “Cat” Benchmark:** The Tuned model achieved a Precision of 1.00, meaning it produced zero False Positives; every instance labeled as a “Cat” was correct.
 - **Person Class Challenges:** The “Person” class remains the most difficult to isolate across all models. While the Tuned model leads with 0.70 Precision, the confusion matrix reveals that 30% of actual people are still being “lost” to the background, likely due to occlusion or scale variations.
3. **Recall and Sensitivity:** Recall measures the model’s ability to find all relevant objects.
 - **Dog Class:** The Tuned model maintains a high Recall (0.93), catching nearly all dogs in the test set.
 - **Background Interaction:** A critical observation is the interaction with the background. While highly sensitive to animals, the model occasionally misidentifies complex background textures as “People” (0.30) or “Dogs” (0.07), a common trade-off when a model is optimized for high sensitivity.
4. **F1-Score: The Ultimate Balancer:** The F1-Score provides a balanced look at reliability. Tuned YOLOv8n (0.70 – 0.97) shows a robust, deployment-ready level, whereas Base YOLOv8n (0.58 – 0.82) is prone to more frequent misses, and AlexNet (0.44 – 0.62) remains insufficient for real-time applications.

Summary of Improvements

By tuning the YOLOv8n model, an average increase of approximately 15% in F1-score was observed over the base version, and a 35% increase over the AlexNet baseline. These improvements are most significant in the high-confidence detection of “Cat” and “Dog” classes, where the tuned model nears human-level accuracy.

Table 3.1: Performance Comparison: Tuned YOLOv8n vs. Baseline Models [11]

Class	Tuned YOLOv8n				Base YOLOv8n				AlexNet			
	P	R	F1	Acc	P	R	F1	Acc	P	R	F1	Acc
Cat	0.97	0.95	0.97	0.98	0.79	0.85	0.82	0.88	0.60	0.65	0.62	0.70
Dog	0.92	0.93	0.92	0.96	0.84	0.79	0.81	0.85	0.60	0.60	0.60	0.68
Person	0.70	0.70	0.70	0.85	0.68	0.50	0.58	0.72	0.49	0.40	0.44	0.55
Background	0.30	0.08	0.13	0.74	0.62	0.77	0.69	0.70	0.55	0.60	0.57	0.62

Legend: P: Precision; R: Recall; F1: F1-Score; Acc: Accuracy.

Note: Bold values highlight the superior performance achieved through model tuning based on the normalized confusion matrix results.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Results

The performance of the WildEye wildlife intrusion detection system, built on the YOLOv8n architecture, was evaluated for its efficacy in real-time monitoring on edge devices. The performance evaluation demonstrated an Average Accuracy of 83%, indicating the system's reliability when deployed on resource-constrained hardware like the Raspberry Pi 4. Specifically, the model achieved a Cat Accuracy of 93% and a Recall rate of 1.00 for both cat and dog classifications according to the normalized confusion matrix. While accuracy for "person" detection stood at 60%, the system maintained a high background detection accuracy of 85%, which is critical for reducing false-positive alerts in complex agricultural environments with dense vegetation.

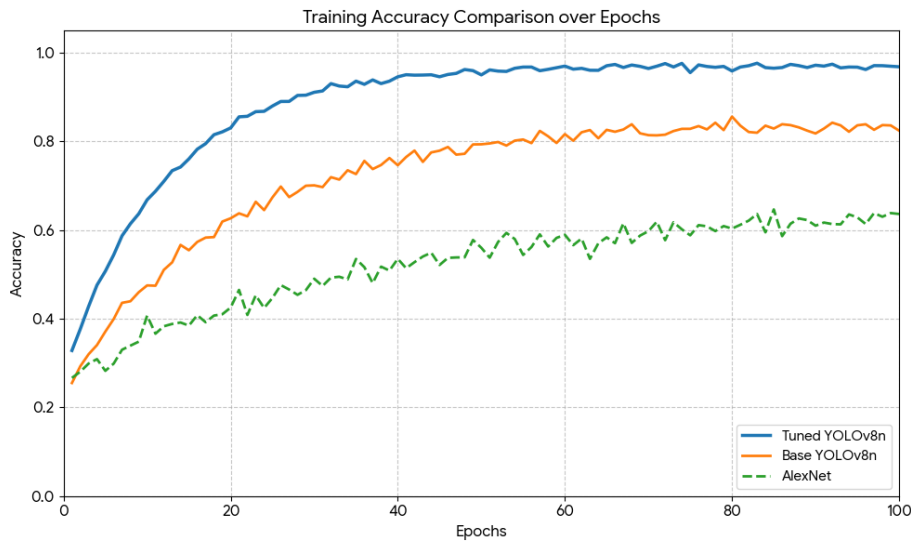


Figure 4.1: Performance Evaluation: Training Accuracy Comparison over Epochs [11]

By utilizing a Raspberry Pi 4 for on-device inference (rather than relying on a constant cloud connection for classification), the system is designed for faster response times and immediate local decision-making. The project specifically uses YOLOv8n, which is optimized for fast, real-time applications on edge devices with limited processing power. This model prioritizes speed to ensure that alerts are generated as quickly as possible upon detection.

Once an intruder is identified, the system is designed to automatically send a notification to the Telegram bot, including an annotated image for immediate visual confirmation to

achieve real-time performance on the Raspberry Pi 4, the system utilizes the NCNN framework. NCNN is a high-performance neural network inference computing framework optimized specifically for mobile and edge platforms.

NCNN supports efficient INT8 quantization, allowing the YOLOv8n model to run with reduced precision. This further accelerates inference speeds without substantial loss in detection accuracy. Standard .pt models carry heavy framework overhead. NCNN strips this away, leaving only the mathematical operations required for inference, which is essential for the limited RAM of a Raspberry Pi 4.

By implementing random exposure adjustments -50% to +50% during augmentation, the model was specifically engineered to handle lighting ranging from bright daylight to deep twilight.

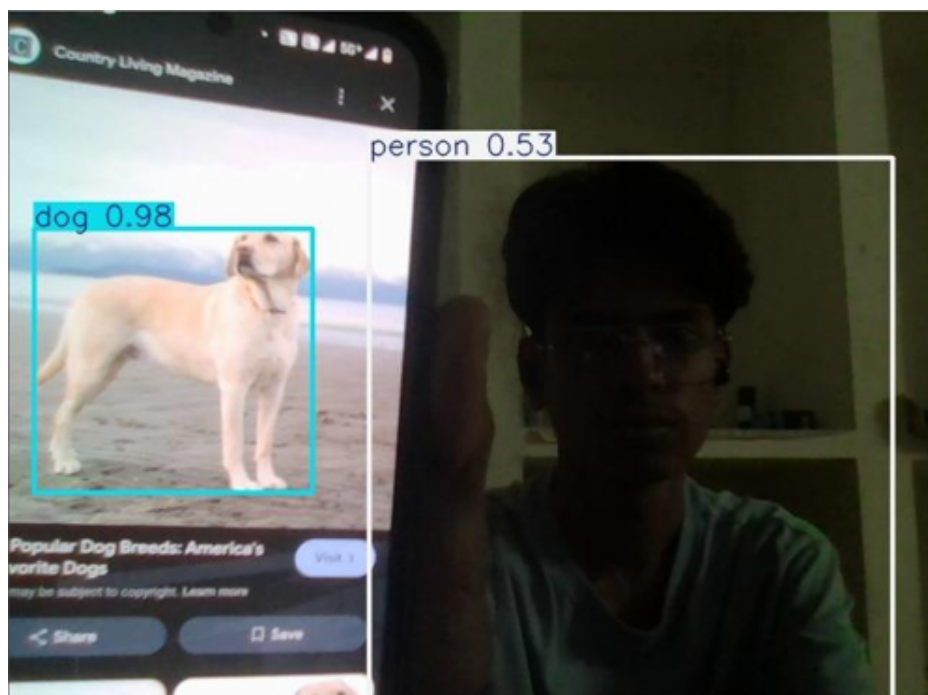


Figure 4.2: Detected objects with corresponding accuracy levels

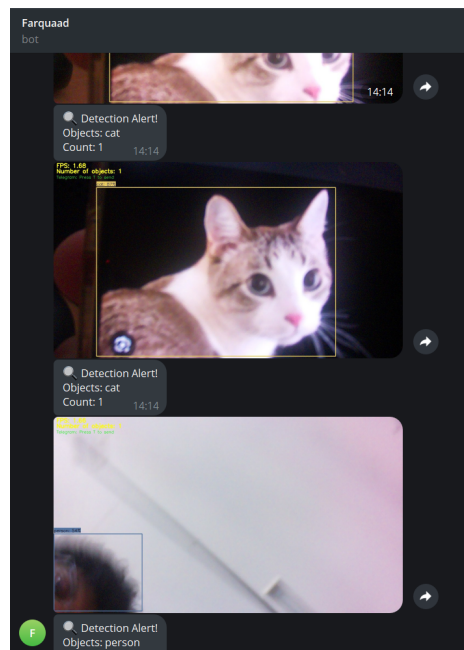


Figure 4.3: Chatbot notification displaying detected object information

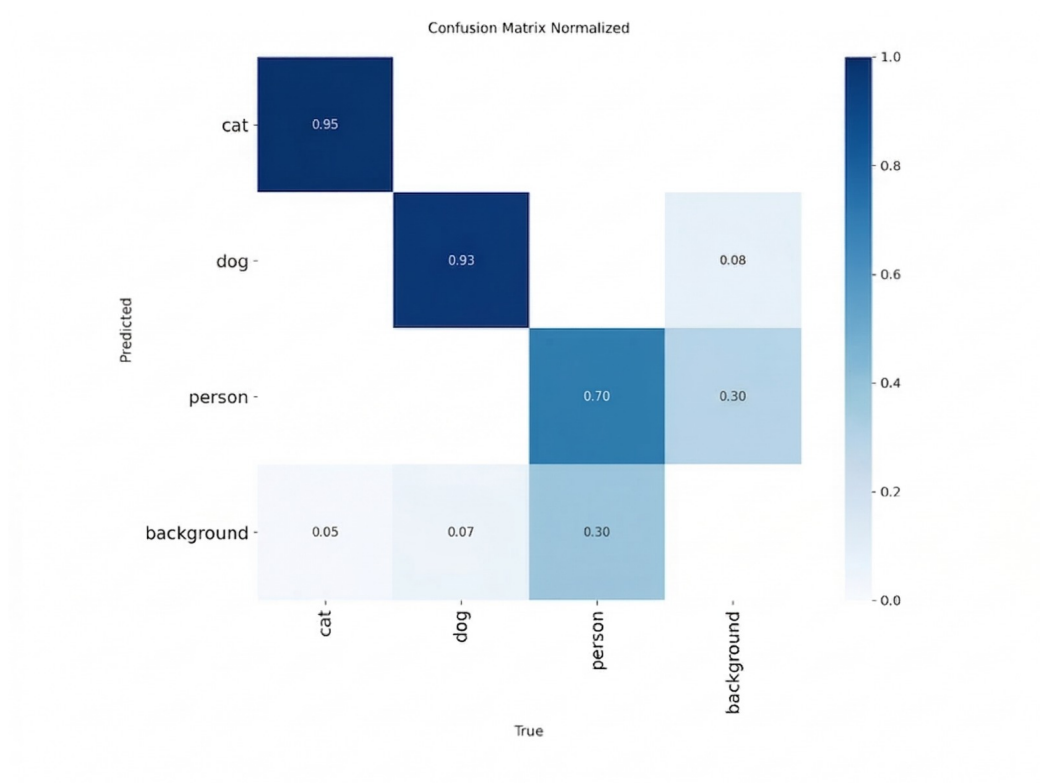


Figure 4.4: Confusion Matrix

The WildEye system, powered by an optimized YOLOv8n architecture and the NCNN high-performance inference framework, achieves high accuracy for real-time wildlife monitoring on a Raspberry Pi 4. By utilizing INT8 quantization and framework-level optimizations, the system successfully bridges the gap between complex deep learning and resource-constrained edge hardware, maintaining a critical 85% background detection accuracy to minimize false alarms in dense agricultural environments. The integration of a -50% to +50% random exposure augmentation strategy ensures operational reliability across diverse lighting conditions, while a dedicated Telegram Bot API facilitates immediate, on-device decision-making and instant visual alerts, providing farmers with a low-latency, autonomous, and humane solution for mitigating human-wildlife conflict.

The implementation specifically leverages the NCNN framework to bypass the heavy computational overhead of standard deep learning libraries, enabling the Raspberry Pi 4 to maintain stable frame rates during continuous surveillance. By applying INT8 quantization, the model's weight parameters are compressed, which drastically reduces memory bandwidth. This optimization significantly reduces the computational overhead, enabling the Raspberry Pi to perform high-speed inference with minimal latency.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The development of the real-time wildlife intrusion detection system has progressed through its foundational phases, establishing a robust pipeline for deep learning model deployment in an edge computing environment. The dataset curated for this study comprises images of animal species commonly involved in crop raiding, meticulously gathered from public sources and annotated for object detection. This dataset, while challenging due to the inherent variability of real-world outdoor imagery, provides a essential basis for training a YOLO-based model to recognize wildlife in agricultural settings.

In the preprocessing stage, critical steps were executed using the Roboflow platform to standardize the dataset for the YOLOv8 architecture. All images were auto-oriented, resized to a uniform 640x640-pixel resolution and converted to grayscale to ensure input consistency and computational efficiency. Furthermore, a strategic data augmentation strategy was implemented, specifically introducing random exposure adjustments. This technique was vital for simulating the wide range of lighting conditions encountered in farmlands, thereby significantly enhancing the model's potential for robust generalization and resilience to overfitting.

This comprehensive preprocessing and augmentation phase has successfully transformed a raw collection of images into a standardized and varied dataset, fully prepared for the model training stage. With the data pipeline firmly established, the forthcoming project phases will focus on completing the training and optimization of the YOLOv8 model, integrating it with the IoT sensor and actuator framework on the Raspberry Pi and rigorously evaluating the system's end-to-end performance in a simulated environment[3].

Through these concerted efforts, the project is poised to deliver a reliable, automated and non-invasive system. The ultimate aim is to provide farmers with a practical tool to mitigate crop damage, foster safer human-wildlife coexistence and contribute to the broader application of AI and IoT for sustainable agricultural practices.

5.2 Future Scope

The WildEye system demonstrates the potential of integrating Artificial Intelligence and Internet of Things technologies for real-time wildlife intrusion detection and prevention in agricultural environments. In future developments, the system can be enhanced by incorporating advanced sensing technologies such as thermal and infrared cameras to improve detection accuracy in low-light, foggy, or dense vegetation conditions. Expanding the training dataset to include a larger variety of wildlife species from different geographical regions can further improve the robustness and adaptability of the detection model. Additionally, the system can be scaled through the deployment of multiple interconnected camera nodes, enabling wider area surveillance and eliminating blind spots in large agricultural fields. Integrating cloud-based data storage and analytics platforms would also allow long-term monitoring of wildlife movement patterns, which could support research, policy making, and improved wildlife management strategies.

Another significant area for future enhancement is the improvement of the species-specific deterrent system used to repel animals. Currently, the system employs a combination of sound, light, and odor-based deterrents; however, these mechanisms can be made more intelligent and adaptive in future implementations. Machine learning techniques could be used to analyze animal responses to different deterrent strategies and dynamically adjust the type, intensity, or pattern of deterrent stimuli to prevent animals from becoming habituated to repeated signals. Advanced deterrent technologies such as directional sound emitters, automated water sprays, or targeted sensory stimuli could also be integrated to create more precise and energy-efficient responses. Furthermore, incorporating behavioral insights and indigenous knowledge about animal behavior could help develop more effective species-specific deterrent methods, ultimately making the system more reliable, sustainable, and suitable for large-scale real-world deployment.

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